

I - Semester Syllabus

1) Applied Mathematics - 1 :

Course Code: BSCP-1	
Course: Practical based on Applied Mathematics-I	
Total Credits: 01	Total Contact Hours: 30 Hrs.
Maximum Marks : 50	
Learning Objectives of the Course:	
i) Apply Taylor's and Maclaurin's series to approximate and analyze functions in engineering problems. ii) Solve real-world limit problems involving indeterminate forms using analytical techniques. iii) Test the behavior of sequences and series for convergence using standard mathematical tools. iv) Compute and apply partial derivatives and Jacobians in multivariable optimization problems. v) Model and solve real-life systems using first- and second-order differential equations.	
Course Outcomes (COs) :	
After completion of the course, students will be able to -	
i) Apply Taylor's and Maclaurin's series to expand functions and solve real-world limit problems involving indeterminate forms. ii) Analyze sequences and series for their convergence or divergence using standard convergence tests such as Comparison, Ratio, and Root tests. iii) Compute first and second-order partial derivatives and evaluate Euler's theorem for homogeneous functions of two variables.	

iv) Evaluate implicit functions, total derivatives, and Jacobians; perform transformations of variables in multivariable functions. v) Determine the maxima and minima of functions of two variables using conditions for critical points and second-order tests. vi) Solve first-order differential equations and apply them to model real-world problems in population growth, electrical circuits, and free-fall motion	
Topics / actual contents of the syllabus	Contact Hours
1. Expand functions using Taylor's and Maclaurin's series 2. Solve practical force-displacement or velocity-time problems where limits result in $0/0$ or ∞/∞ forms. 3. Analyze the behavior of Sequences and Their Limits 4. Test the Convergence and Divergence of Infinite Series 5. Compute first-order and second-order partial derivatives Partial Derivatives of Functions 6. Verify Euler's Theorem on Homogeneous Functions 7. Differentiate functions implicitly and compute total derivatives Objective:- 8. Compute Jacobians and apply transformation of variables 9. Find the Maxima and Minima of Functions of Two Variables 10. Solving First-Order Differential Equations 11. Solve Exact Differential Equations 12. Solve linear differential equations with a population growth model 13. Analyze simple electrical circuits using differential equations 14. Apply differential equations to solve a free-fall motion equation	30 Hrs
Students should undertake at least 08 to 10 experiments during the semester from above list	

2) Engineering Physics :

Course Code : BSCT-2A

Course: Engineering Physics

Total Credits 03

Total Contact Hours : 45 Hrs.

Maximum Marks : 100

Learning Objectives of the Course:

- i) To let the engineering undergraduates study physical properties, concepts and physical quantities required for the solution of complex engineering problems
- ii) To make the engineering undergraduates learn basic principles of Physics and laws of scientific investigation to identify, formulate and analyse complex engineering problems
- iii) To equip engineering undergraduates with competencies of scientific methods required in engineering career by upgrading skills on the basis of learning achieved from physical science perspectives.
- iv) To engage engineering undergraduates extensively in scientific investigation for interdisciplinary graduate programs and a wide variety of other lifelong learning opportunities.

Course Outcomes (COs) :

After completion of the course, students will be able to -

- i) Recall basics of Optics, Sound and Modern Physics
- ii) Explain phenomenon in Optics, Sound and Modern Physics
- iii) Apply concepts of Optics, Sound and Modern Physics to solve complex engineering problems
- iv) Analyse the concepts of Optics, Sound and Modern Physics
- v) Interpret the characteristics of Solar cell, Zener diode, Transistor, Planks constant curve, GM Counter curve
- vi) Experiment with Reverberation Time, Energy Band Gap, Wavelength of Light, Dielectric Constant, Specific Rotation, Wavelength of Ultrasonic waves, divergence of the laser beam, and hall coefficient

ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	A:Optics The wave equation, Introduction to electromagnetic waves and electromagnetic spectrum, Newton's ring, Michelson interferometer, Applications of interference Diffraction of light, diffraction grating, resolving power of grating, Application of diffraction grating in spectroscopic devices B:Ultrasonics	15 Hrs

	Properties, Production of ultrasonic waves by piezo-electric and magnetostriction generator, engineering applications of ultrasonic waves.	
II	A:X-Rays Basics of X-Rays, Production and Detection of X-Rays, Continuous and characteristics spectrum, Bragg's law of X-ray diffraction, Bragg's spectrometer, Intensity of diffracted Beams, Particle Size Determination by XRD, Precise Lattice Parameter Determination B:Modern Physics Black body radiation, Planck's law, Photoelectric effect, Wave particle duality, De-Broglie's concept of matter wave, Davisson-Germer experiment, Scanning tunneling microscope, Time-dependent and time-independent Schrodinger equation for wave function, Quantum computing.	15 Hrs
III	A:Introduction to solids Superconductivity: Superconductivity, effect of temperature and magnetic fields, Meissner effect, type I and II superconductors, BCS theory, Applications. Free electron theory of metals, Fermi level, density of states, Application to white dwarfs and neutron stars, Bloch's theorem for particles in a periodic potential, Kronig-Penney model and origin of energy bands B:Laser Einstein's theory of matter radiation interaction and A and B coefficients, Properties of laser, spontaneous and stimulated emission, ruby laser, He-Ne laser, CO ₂ laser and semiconductor Laser, applications of lasers in science, engineering and medicine. C:Fiber Technology Propagation of light through optical fiber, acceptance angle and cone numerical aperture, Single and Multi-Mode Fibers, applications, sensors	15 Hrs
Text Books: 1. M. N. Avadhanulu P. G. Kshirsagar , " A Text book of Engineering Physics ", 7 th Edition, S. Chand & Co.		

2. R. K. Gaur S. L. Gupta ., “ A Text book of Engineering Physics ”, 3rd Edition, Dhanpat Rai.

Reference Books:

1. David Halliday, Jearl Walker, and Robert Resnick “ Fundamentals of Physics”, 6th Edition , Wiley
2. B. D. Cullity , “ Elements of X-ray Diffraction ”, 1st Edition , Addison-Wesley Metallurgy Series.

3) P-Programming language :

Course Code : ESCT-1 Course: Principles of Programming Language Total Credits 03 Total Contact Hours : 45 Hrs. Maximum Marks : 100		
Learning Objectives of the Course: i) To introduce the various programming paradigms and evolution of programming languages. ii) To introduce the principles and techniques involved in design and implementation of modern programming languages. iii) To introduce the notations to describe the syntax and semantics of programming languages. iv) To develop logics which will help them to create programs, applications in C		
Course Outcomes (COs) : After completion of the course, students will be able to - i) Identify various programming languages, their features, and execution models including compilers and interpreters. ii) Describe data types, variables, expressions, and control structures in C programming. iii) Apply conditional statements, loops, and functions to solve basic computational problems using C. iv) Analyze problems using arrays, strings, and structured data types to implement modular C programs. v) Develop structured and modular programs using functions, recursion, and pointer-based parameter passing. vi) Construct applications by integrating arrays, strings, structures, and functions to handle real-world problems.		
ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	A:Introduction to programming languages Introduction to computers, input and output devices, programming languages, programming domains, language evaluation criteria, influences on language design, language categories. Programming Paradigms-Imperative, Functional Programming language, assembler, compiler, interpreter, Syntax and Semantics, Algorithm, Flowchart, pseudocode B:Names, Bindings, and Scopes Introduction, names, variables and identifiers, concept of binding, scope and lifetime	15 Hrs.

	Data types: Introduction, primitive, character, string types, user defined ordinal types, Data types in c Expressions and Statements: Arithmetic expressions, type conversions, relational and Boolean expressions, short- circuit evaluation, assignment statements, mixed mode assignment, Expression and statements in C	
II	A:Control Structure and Loops in C Control Structures – introduction, selection statements, iterative statements, unconditional branching, guarded commands. If-else, conditional operators, switch and break, nested conditional branching statements, loops: do while, while, for, Nested loops, break and continue B:Subprograms And Blocks Fundamentals of Subprograms, difference between procedures and functions. Functions in C: Introduction to functions, Function definition, function declaration, function call, return statement, passing parameters to functions, scope of variables, recursive functions, concept of pointers, call by value, call by reference	15 Hrs.
III	A:Arrays in C Declaration of arrays, accessing the elements of an array, storing values in arrays, Operations on arrays, Passing arrays to functions, two dimensional arrays, operations on two-dimensional arrays String: String declaration, initialization, string manipulation with/without using library functions B:Structures in C Introduction, defining a structure, declaring structure variables, accessing structure members, structure initialization, array of structures, Union and Enumeration	15 Hrs.
Text Books: I Yashavant P. Kanetkar , “Let Us C” , 16 th Edition, BPB Publications .		
Reference Books:		

1. Robert .W. Sebesta , “ Concepts of Programming Languages ”, 10th Edition, Pearson Education
2. D. A. Watt , “ Programming Language Design Concepts ”, 1st Edition, Wiley India Edition
3. E. Balaguruswamy, “ Programming in ANSI C ”, 8th Edition , McGraw Hill Publications

4) Basic of Electrical Engineering :

Course Code : ESCT-2

Course: Basics of Electrical Engineering

Total Credits 03

Total Contact Hours : 45 Hrs.

Maximum Marks : 100

Learning Objectives of the Course:

- i) The objective is to present essential principles, laws, and theorems related to electrical systems.
- ii) The aim is to provide a foundational understanding of electrical fundamentals, encompassing current, voltage, power, energy, and frequency.
- iii) The goal is to familiarize students with key parameters like resistance, inductance, capacitance, magnetic circuits, AC and DC circuits

Course Outcomes (COs) :

After completion of the course, students will be able to -

- i) Recall fundamental electrical concepts such as resistance, capacitance, inductance, and electric potential.
- ii) Explain the behavior of DC and AC circuits using Kirchhoff's laws, Ohm's law, and electromagnetic principles
- iii) Apply network theorems like Superposition, Thevenin's, and Maximum Power Transfer to analyze simple electrical circuits.
- iv) Analyze single-phase R-L, R-C, and RLC circuits using phasor diagrams and evaluate power factor and resonance conditions.
- v) Evaluate performance of DC and AC circuits using loop/nodal analysis and source transformation techniques.
- vi) Compare electrical and magnetic circuits and interpret the functioning of balanced three-phase systems using phasor diagrams.

ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	A: Introduction Effect of temperature on resistance, Resistance coefficient, Derivation for relations between α_1 , α_2 , Ohm's Law, Work, Power energy, and the relationship between Thermal, and electrical units.) Series and parallel combination of resistors (simple problems based on above) D.C. Networks, Kirchhoff's laws, Ideal & Practical sources, source conversions Loop and nodal analysis, Superposition Theorem, (with two sources, no current source in numerical) Thevenin's, & Maximum power transfer theorem (No super node or mesh) (Numerical)	15 Hrs.

II	A: Electrostatics: Electric potential, potential difference, electric potential, Conductors and insulators, capacitance, a combination of capacitors in series and in parallel, energy stored in a capacitor.(No Numerical) B: Magnetic Circuits: BH Curve, Inductance, Induced emf, its types static and dynamic – Self & mutual induced emf, series magnetic circuits, coefficient of coupling (Derivation) comparison of Electrical and magnetic circuits) (Numerical on single reluctance magnetic circuits)	15 Hrs.
III	A: A.C. Circuits Sinusoidal voltage and current waveforms, RMS and average value. Form factor and crest factor R-L, R-C, RLC series circuits, Phasor diagram, power and power factor, series resonance.(Simple numerical) B: Three Phase Balanced Systems, Three phase voltage generation and waveform star and delta balanced systems, Relationship between phase and line quantities, Phasor diagram, power in a three-phase circuit(No numerical)	15 Hrs.
Text Books: 1 B.L.Thereja, “ Electrical Technology Vol.I&II ”,24th Edition S.Chand. 2 B.L.Thereja and A.K.Thereja, “ABC of Electrical Engineering”,1st Edition S.Chand 3 J.B. Gupta , “ Basic Electrical Engineering ”,14th Edition,S.K. Kataria& sons. 4 V.K.Mehta , “ Basic Electrical Engineering ”,2nd Edition, S.Chand		
Reference Books: 1. E.Huges , “ Basic Electrical Engineering ”,10th Edition, Mc-Graw Hill,		

5) Professional Communication skill:

Course Code : AECT-1 Course: Professional Communication Skills Total Credits 01 Total Contact Hours : 15 Hrs. Maximum Marks : 50		
Learning Objectives of the Course: i) To apply English parts of speech in day to day communication. ii) To apply English Tenses in situational communication iii) To apply transformation of sentences in professional communication iv) To pronounce English words and sentences accurately v) To communicate in English effectively by using updated vocabulary		
Course Outcomes (COs) : After completion of the course, students will be able to - i) Recall and define basic grammatical terms such as parts of speech, tenses, and sentence types. ii) Explain the use of tenses with timelines and differentiate between active and passive voice in various sentence structures. iii) Transform sentences between different voices, types (simple, compound, complex), and speech (direct–indirect). iv) Identify and analyze conditional clauses and errors in pronunciation, stress, and articulation in spoken English. v) Evaluate the effectiveness of vocabulary usage and pronunciation through peer discussion and feedback. vi) Develop vocabulary enhancement activities and construct grammatically correct and phonetically sound sentences for various contexts.		
ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	Parts of Speech Tenses with Timeline Transformation of sentences: Simple, compound and complex Active and passive voice Conditional Clauses	5 Hrs
II	Phonetics and problems in learning and using pronunciation, Vowel sounds & Consonant Sounds, Articulation of Sounds	5 Hrs

III	Types of Vocabulary	5 Hrs
	Basic techniques to Enhance Vocabulary	
	Vocabulary Enhancing Activities	
Text Books:		
1 A. K. Jain, Pravin, S. R. Bhatia, A. M. Sheikh, "Professional Communication Skills", S. Chand & Company Ltd. 2 Urmila Rai, S. M. Rai, "Business Communication", 9th Edition , Himalya Publishing House 3 Wren and Martin, " English Grammar and Composition ", 1st Edition , S. Chand Publications		
Reference Books:		
1 Adrian Budday, Ron Ludlow and Fergus' Panton, "The Essence of Effective Communication", Prentice Hall of India-Private Ltd. 2 Meenakshi Raman & Sangeeta Sharma., "Technical Communication- Principles and Practice", 2nd Edition Oxford University Press. 3 J. Sethi, P.V. Dharmatma "A course in Phonetics & Spoken English.", 2nd Edition PHI Publication 4 Sunita Mishra, C. Murli Krishna "Communication Skills for Engineers.", 2nd Edition Pearson Education 5 Douglas Biber, Geoffrey Leech "Grammar of Spoken and Written English.", 1st Edition Longman		

6) Design Thinking :

Vocational And Skill Enhancement Course (VSEC)		
Course Code : VSECT-1 Course : Design Thinking Total Credits01 Total Contact Hours :15 Hrs. Maximum Marks : 50		
Learning Objectives of the Course: i) To familiarize students with the fundamental concepts and stages of the Design Thinking process. ii) To enable students to generate innovative solutions and develop strategic plans for implementing these solutions in real-world engineering contexts. iii) To equip students with the ability to create and test low-fidelity prototypes. Course Outcomes (COs) : After completion of the course, students will be able to - i) Understand the fundamentals of design thinking, including its history, benefits, and application in engineering problem-solving. ii) Apply empathy techniques to identify user needs and construct empathy maps through research and observation. iii) Formulate clear and actionable problem statements using structured problem identification methods. iv) Use ideation techniques like brainstorming, mind mapping, and brainwriting to generate innovative solutions. v) Develop low-fidelity and high-fidelity prototypes using suitable tools and materials to represent potential solutions. vi) Analyze user feedback and apply iterative testing methods to refine and improve design prototypes.		
Module No.	Topics / actual contents of the syllabus	Contact Hours
I	Introduction to Design Thinking: Introduction, history, and benefits, design challenges and opportunities. Design Thinking process: General design process, Scope for design, Design thinking process, Relevance to engineering Empathy: Identifying customer Needs, Techniques for user research, building empathy map	5 Hrs.
II	Defining problem: Methods to identify core problems, crafting problem statements	5 Hrs.

	Ideation techniques: Brainstorming, brainwriting, mind mapping, out-of-box thinking, Brainstorming techniques for generating ideas, Idea selection and evaluation methods.	
III	Prototyping Definition and purpose of prototyping, Difference between low-fidelity and high-fidelity prototypes, Benefits of prototyping in iterative design processes, Tools and material used for prototyping. Testing: Definition and purpose of testing in the design process, Importance of user feedback in refining prototypes, Testing as an iterative process	5 Hrs.
ReferenceBooks: 1. C.L. DymP.Little, "Engineering Design: A Project Based Introduction", 4th Edition, WileyPublication 2. Karl Ulrich., "Project Design & Development", 5th Edition McGraw Hill Publication. 3. Green, W., & Jordan, P. W., "Human factors in product design: current practice and future trends..", 1st EditionCRC Press 1999		

Course Code : CCT-1

Course: Yoga

Total Credits 01

Total Contact Hours : 15 Hrs.

Maximum Marks : 50

Learning Objectives of the Course:

- i) To identify common stressors inherent in today's global marketplace.
- ii) To develop an understanding of the impact of stress on physiological, emotional and cognitive processes.
- iii) To learn to manage the stress through art of Yoga.

Course Outcomes (COs) :

After completion of the course, students will be able to -

- i) Define the concepts of health, stress, and mental well-being, and explain the yogic and physiological perspectives on mental health and stress.
- ii) Demonstrate understanding of the symptoms, causes, and consequences of stress and apply yogic concepts and techniques to manage stress.
- iii) Analyze the physiological impact of stress on the Autonomic Nervous System, Endocrine system, and Hypothalamus and distinguish between psychic, psychosomatic, somatic, and organic stress responses.
- iv) Evaluate the role of meditation and pranayama in regulating stress and improving overall health from a physiological and psychological perspective.
- v) Perform and design stress control techniques such as sitting/walking meditation, progressive muscle relaxation, gentle stretches, and massage for personal and therapeutic use.
- vi) Examine the preventive and curative effects of yoga on stress-related disorders such as hypertension, asthma, diabetes, and anxiety.

ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	Mental Health: Meaning and importance; Yogic Perspective of Mental Health, Indicators of Mental Health, Stress: Meaning and Definition; Symptoms, Causes and Consequences of Stress, Meaning of Management- Stress Management, Stress in Modern Culture & Society. Concept of Stress according to Yoga, Assessing your Stress & Building Resilience	5 Hrs
II	Physiology of Stress on: Autonomic Nervous System (ANS), Endocrine System, Hypothalamus, Mechanism of Stress related diseases: Psychic, Psychosomatic, Somatic and Organic phase. Role of Meditation & Pranayama	5 Hrs

	on stress- physiological aspect of meditation, constant stress & strain, anxiety.	
III	Meaning and definition of Health: various dimensions of health (Physical, Mental, Social and Spiritual) Yoga and Health- Yoga as therapy, Physical fitness. Stress control exercise- Sitting meditation, Walking meditation, Progressive muscular relaxation, Gentle stretches and Massage. Preventive and curative effects of Yoga on stress related disorders: Hypertension, Heart problems, Bronchial Asthma, Peptic Ulcer, Diabetes Mellitus, Arthritis, Anxiety Neurosis and Headache	5 Hrs
Text Books (If Any): 1 H.R.Nagendra, and R. Nagarathana , " Yoga perspective in stress management ", Swami Vivekananda Yoga Prakashana 2004 2 H.R.Nagendra, and R. Nagarathana , "Yoga practices for anxiety & depression", Swami Sukhabodhanandha Yoga Prakashna 2004 3 K.N.Udupa , " Stress management by Yoga ", Motilal Banaridass Publishers Private Limited 1996,		
Reference Books: 1 Linda Wasmer Andrews , " Stress Control for peace of Mind ", Main Street 2005. 2 Vimla Lalvani , " Yoga for stress ", Hamlyn 1998. 3 B.K.S.Lyengar " Light on Yoga .", Thorson		